Named Entity Recognition (NER) on Business Document – 3rd April

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NER → Named Entity recognition.

↳ NLP task used to identify and classify important entities in the text

Extract real world entities.

- Person. (eg) name.
- Location
- Organization
- Email, skill etc.

NER is a fundamentally a Token classification problem.

Why?

- Text is first broken into token (words | subwords) using tokenizer.
- Each token is then assigned a label (entity types).

Process flow.

Input sentence (text) → Tokenization → Token wise classification → Entity extraction.

Eg. Bill Gates is a founder of Microsoft.

Token	Entity type	Grammatical role
Bill	Person.	Noun.
Gate	Person.	Noun.
is	Other | Object → O	Verb.
a	Other | object → O	Article
founder	Designation.	Noun.
of	Other.	preposition
Microsoft	Organization	Proper noun

Application

NER is widely used in real-world system.

- Resume parsing.

  Extract
  - Name
  - skills.
  - Experience
  - Education
  - Email

- Banking | Financial Document

Cards

Account holder name

Transaction detail

Location

Token classification

- NER
- POS Tagging (Part of Speech)
- Chunking.

Pipeline in advance NLP system

- Text preprocessing
- Tokenization
- POS Tagging.
- NER.
- Chunking.
- Task (Chatbot, summarization etc).